

Worksheet

Computer Network - Week 5 - UDP

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Activity 1

Screenshots

The screenshot displays the Wireshark network traffic analysis tool. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for file operations, capture control, and analysis. The main display area is divided into three panes: the packet list, packet details, and packet bytes.

The packet list pane shows a list of captured packets. The selected packet is packet 291, which is a DNS Standard query response from 10.6.47.254 to 10.6.47.43. The details pane shows the structure of the User Datagram Protocol (UDP) and Domain Name System (DNS) message. The packet bytes pane shows the raw data of the packet in hexadecimal and ASCII.

The packet list pane shows the following packets:

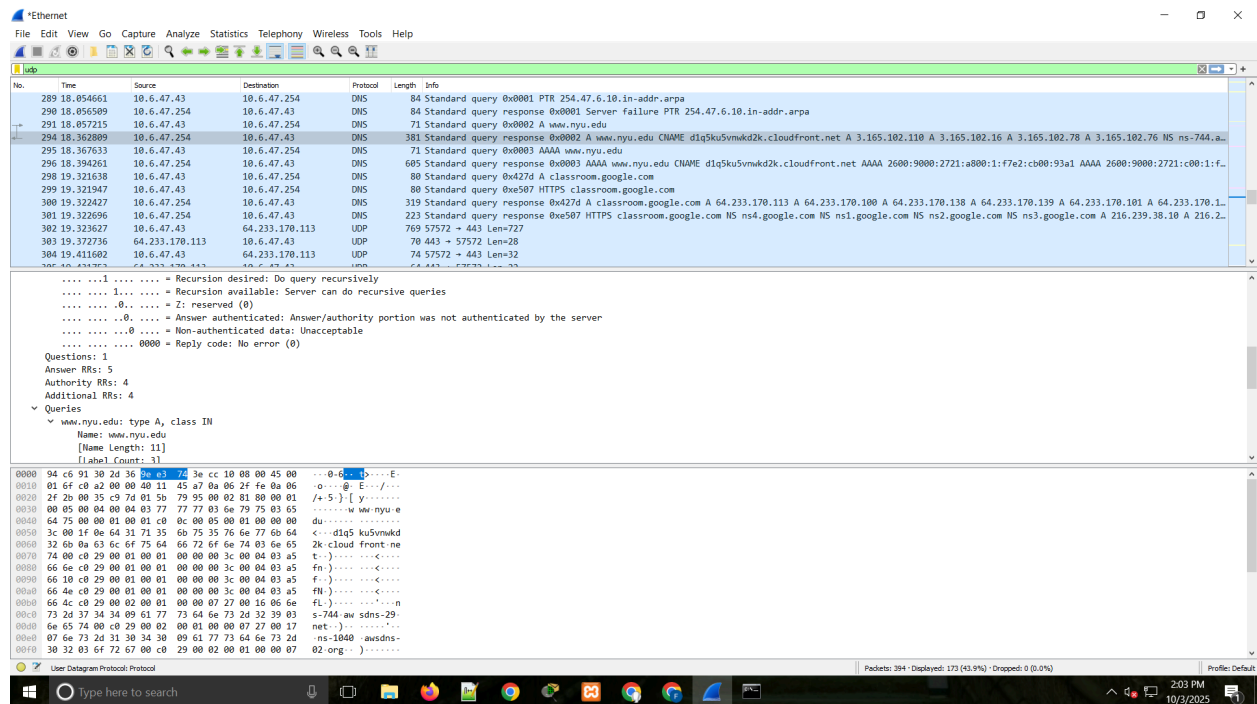
No.	Time	Source	Destination	Protocol	Length	Info
289	18.054661	10.6.47.43	10.6.47.254	DNS	84	Standard query 0x0001 PTR 254.47.6.10.in-addr.arpa
290	18.056509	10.6.47.254	10.6.47.43	DNS	84	Standard query response 0x0001 Server failure PTR 254.47.6.10.in-addr.arpa
291	18.057215	10.6.47.43	10.6.47.254	DNS	71	Standard query 0x0002 A www.nyu.edu
294	18.362809	10.6.47.254	10.6.47.43	DNS	381	Standard query response 0x0002 A www.nyu.edu CNAME d1q5k5vmkd2k.cloudfront.net A 3.165.102.110 A 3.165.102.16 A 3.165.102.70 A 3.165.102.76 NS ns-744.a...
295	18.367633	10.6.47.43	10.6.47.254	DNS	71	Standard query 0x0003 AAAA www.nyu.edu
296	18.394261	10.6.47.254	10.6.47.43	DNS	605	Standard query response 0x0003 AAAA www.nyu.edu CNAME d1q5k5vmkd2k.cloudfront.net AAAA 2600:9000:2721:a800:1:f7e2:c800:93a1 AAAA 2600:9000:2721:c00:1:f...
298	19.321638	10.6.47.43	10.6.47.254	DNS	80	Standard query 0x427d A classroom.google.com
299	19.321947	10.6.47.43	10.6.47.254	DNS	80	Standard query 0xe507 HTTPS classroom.google.com
300	19.322427	10.6.47.254	10.6.47.43	DNS	319	Standard query response 0x427d A classroom.google.com A 64.233.170.113 A 64.233.170.100 A 64.233.170.130 A 64.233.170.139 A 64.233.170.101 A 64.233.170.1...
301	19.322696	10.6.47.254	10.6.47.43	DNS	223	Standard query response 0xe507 HTTPS classroom.google.com NS ns4.google.com NS ns1.google.com NS ns2.google.com NS ns3.google.com A 216.239.38.10 A 216.2...
302	19.323627	10.6.47.43	64.233.170.113	UDP	769	57572 → 443 Len=727
303	19.372736	64.233.170.113	10.6.47.43	UDP	70	443 → 57572 Len=28
304	19.411602	10.6.47.43	64.233.170.113	UDP	74	57572 → 443 Len=32

The packet details pane shows the following information:

- User Datagram Protocol, Src Port: 51581, Dst Port: 53
- Source Port: 51581
- Destination Port: 53
- Length: 37
- Checksum: 0x000d [unverified]
- [Checksum Status: Unverified]
- [Stream index: 18]
- [Timestamps]
- [Time since first frame: 0.000000000 seconds]
- [Time since previous frame: 0.000000000 seconds]
- UDP payload (29 bytes)
- Domain Name System (query)
- Transaction ID: 0x0002
- Flags: 0x0100 Standard query
- 0
- 0x0000 0a e3 7d 3e cc 10 94 c6 91 30 2d 36 00 00 45 00 - 13-... 0-6-E
- 0x0010 00 39 69 b2 00 00 80 11 5d cd 0a 06 2f 2b 0a 06 - 91-...]-.../...
- 0x0020 2f fe c9 7d 00 35 00 25 60 0d 00 02 01 00 00 01 - /-)-5-%-.....
- 0x0030 00 00 00 00 00 00 03 77 77 77 03 6e 79 75 03 65 -w ww.nyu.e
- 0x0040 64 75 00 00 01 00 01 - du-xx--

The packet bytes pane shows the raw data of the packet in hexadecimal and ASCII.

Request segment



Response segment

Answers

- Select the first UDP segment in your trace.
 - What is the **packet number** of this segment in the trace file?
 - What type of **application-layer payload** or **protocol message** is being carried in this UDP segment?
 - How many **fields** are there in the UDP header?
 - What are the **names** of these fields?

The packet number is 291
 The protocol message is DNS
 There are 5 fields
 source port, destination port, length, and checksum

- By inspecting the displayed information in Wireshark's packet content field for this packet (or by consulting the textbook), what is the **length** (in bytes) of each of the UDP header fields?

We can see the length in the bottom of the app
Source port : 2 bytes
Destination port : 2 bytes
Length : 2 bytes
Checksum : 2 bytes

3. What is the **Length field** referring to?. Verify your claim with your captured UDP packet.

It stand for UDP length = IP length + IP header's length = 29 + 8 = 37 bytes

4. Examine the pair of UDP packets in which your host sends the first UDP packet and the second UDP packet is a reply to this first UDP packet.
- What is the **packet number** of the **first** of these two UDP segments in the trace file?
 - What is the **value** in the **source port field** in this UDP segment?
 - What is the **value** in the **destination port field** in this UDP segment?
 - What is the **packet number** of the **second** of these two UDP segments in the trace file?
 - What is the **value** in the **source port field** in this second UDP segment?
 - What is the **value** in the **destination port field** in this second UDP segment?
 - Describe the **relationship** between the **port numbers** in the two packets.

Packet number for the first UDP segment = 51581

Source port field = 51581

Destination port field = 53

Packet number for the second UDP segment = 53

Source port field(2) = 53

Destination port field(2) = 51581

The request segment has it own source port field, the destination port field is the source port field of the response segment, also the the destination port field of the response segment is the source port field of the request segment, i think they use it as an identification for these two segments for communicating